

unit 3SHA Algo. [Secure Hash value]

It

output $\rightarrow$ 160 bit. $\Rightarrow$ At the end of message add message original length

Variable length IP $\rightarrow$ SHA-1 $\rightarrow$ Hash value [160 bits]

$\rightarrow$ In MD5 - length of O/P $\rightarrow$ 128 bits
 In SHA $\rightarrow$ length of O/P $\rightarrow$ 160 bits.

 $\rightarrow$ working-

1000 bits

 $\rightarrow$ Padding - (MD5) [64 bit]

$$448 + 64 \text{ bits} = 512$$

Appending 64 bit rep [512 mult p]

512 512 512 512
 $\rightarrow$ Divide the IP into 512 bit block $\rightarrow$ initialize (5) chaining variable

(A, B, C, D & E)

 $\rightarrow$ Process block $\rightarrow$ copy corresponding block

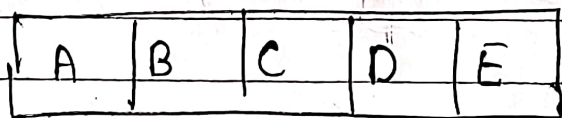
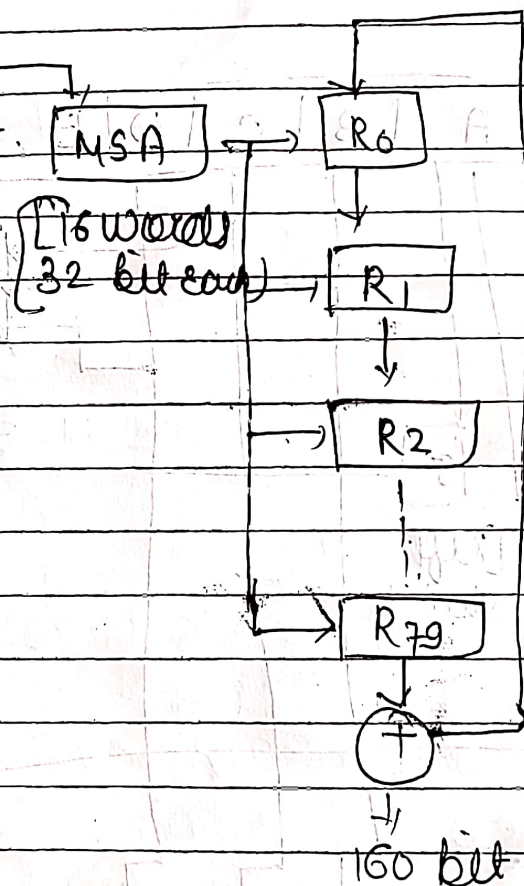
→ Divide into no. of 512 bit blocks.
(16-32)

→ Four rounds (each rounds = 20 step)

$$M(M_1, M_2, M_3 \dots M_4)$$
$$448 + 64 \text{ bit} = 512 \text{ bit}$$

↓ ↘
Actual around
bit End.

10000



A - H₀ (67452301)

$$B-1) H_2 (EFCD) \rightarrow + 9$$

C-1 H₂

$$D \rightarrow H_2$$
$$H \rightarrow H$$

K15

R16

$$\frac{k}{8t}$$

K.

$$0 \leq t \leq 1$$
 $0.0 < t_k$
$$40 \leq t \leq 5$$

605 E 57

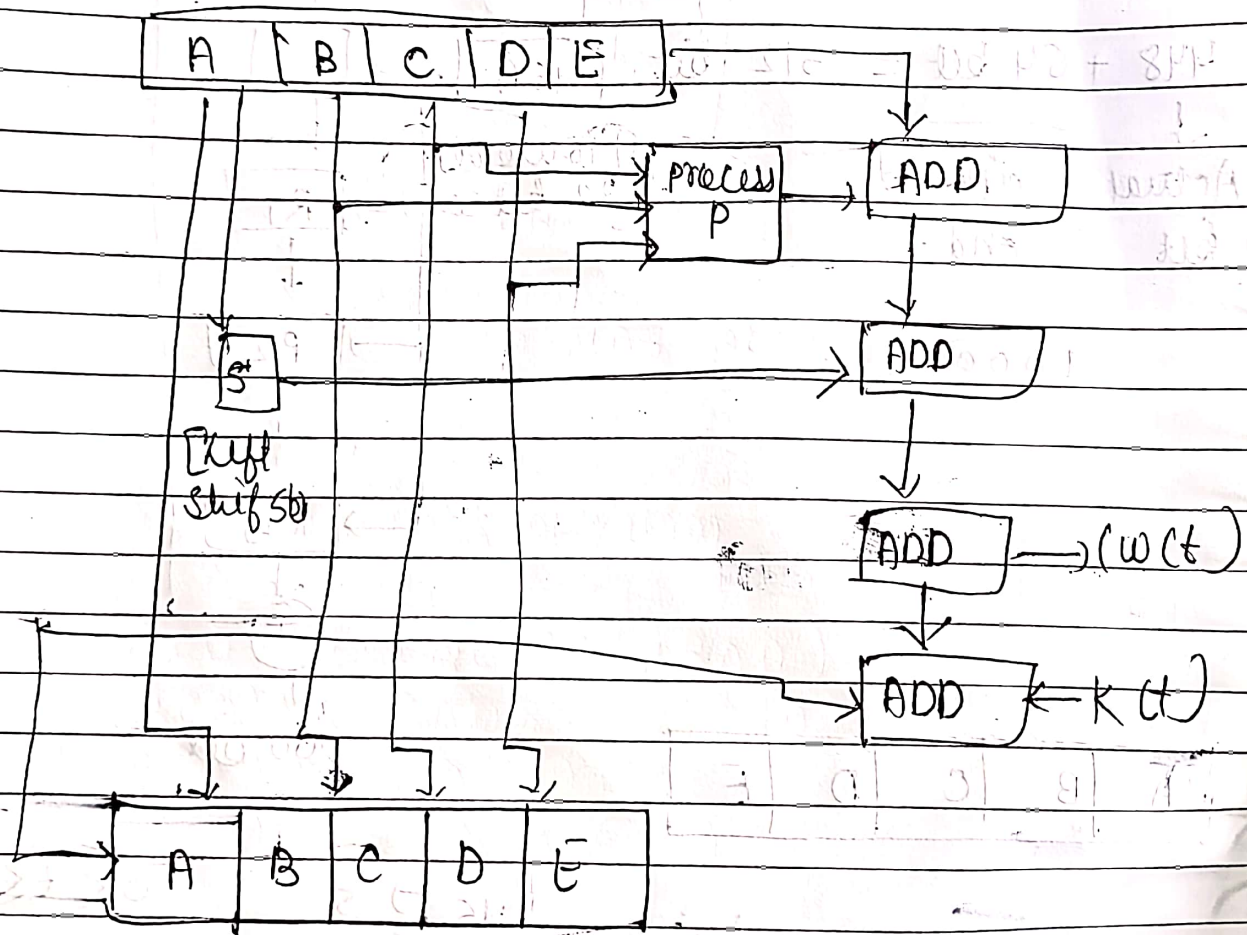
→ Example

→ Password storage

→ Digital signatures

→ Data integrity

→ Round function



$$ABCDE = E + \text{Process } P + S(a) + w(c) + K(c)$$

eg

$$512 \times 2 \rightarrow 1024$$

$$512 \times 3 \rightarrow 1536$$

$$512 \times 4 \rightarrow 2048$$

$$1000 + 64$$

$$1064$$

$$1000 + 472 + 64$$

$$1536$$

$$64$$

$$1472$$

$w(t) \rightarrow$ is the expanded message word of round t
 $K(t) \rightarrow$ is the round of constant t

$\rightarrow$ SHA consists of four rounds, each round containing 40 round each.

$\Rightarrow$ MDS Algo $\rightarrow$ [Message digest Algo.]

1) Append padding bits.

2) Append 64 bits represent [Exact multiple of 512]

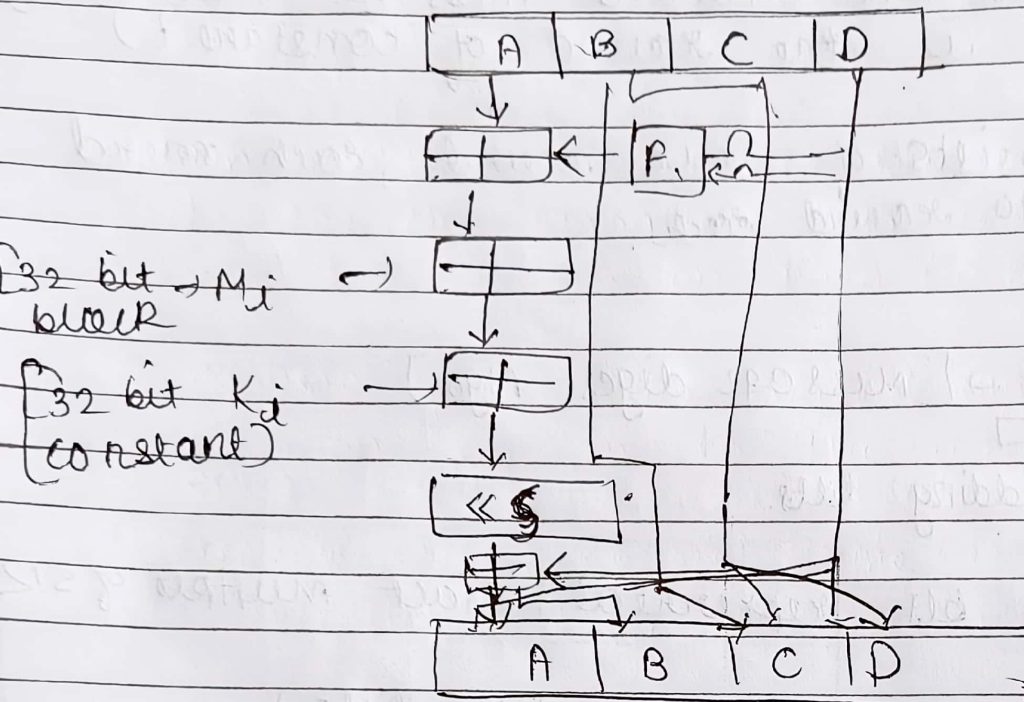
3) Initialize MDS Buffers $\rightarrow$ [32 bits]
 out: 128 bits

4) Process each block.

$$1000 + 472 + 64$$

1536

512	512	512
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Round $\rightarrow 4$

one Round $\rightarrow 16$ operation.

A $\rightarrow 0x64253$

$$F(B, C, D) \rightarrow (B \wedge C) \vee (\neg B \wedge D)$$

$$G(B, C, D) \rightarrow (B \wedge D) \vee (C \wedge \neg D)$$

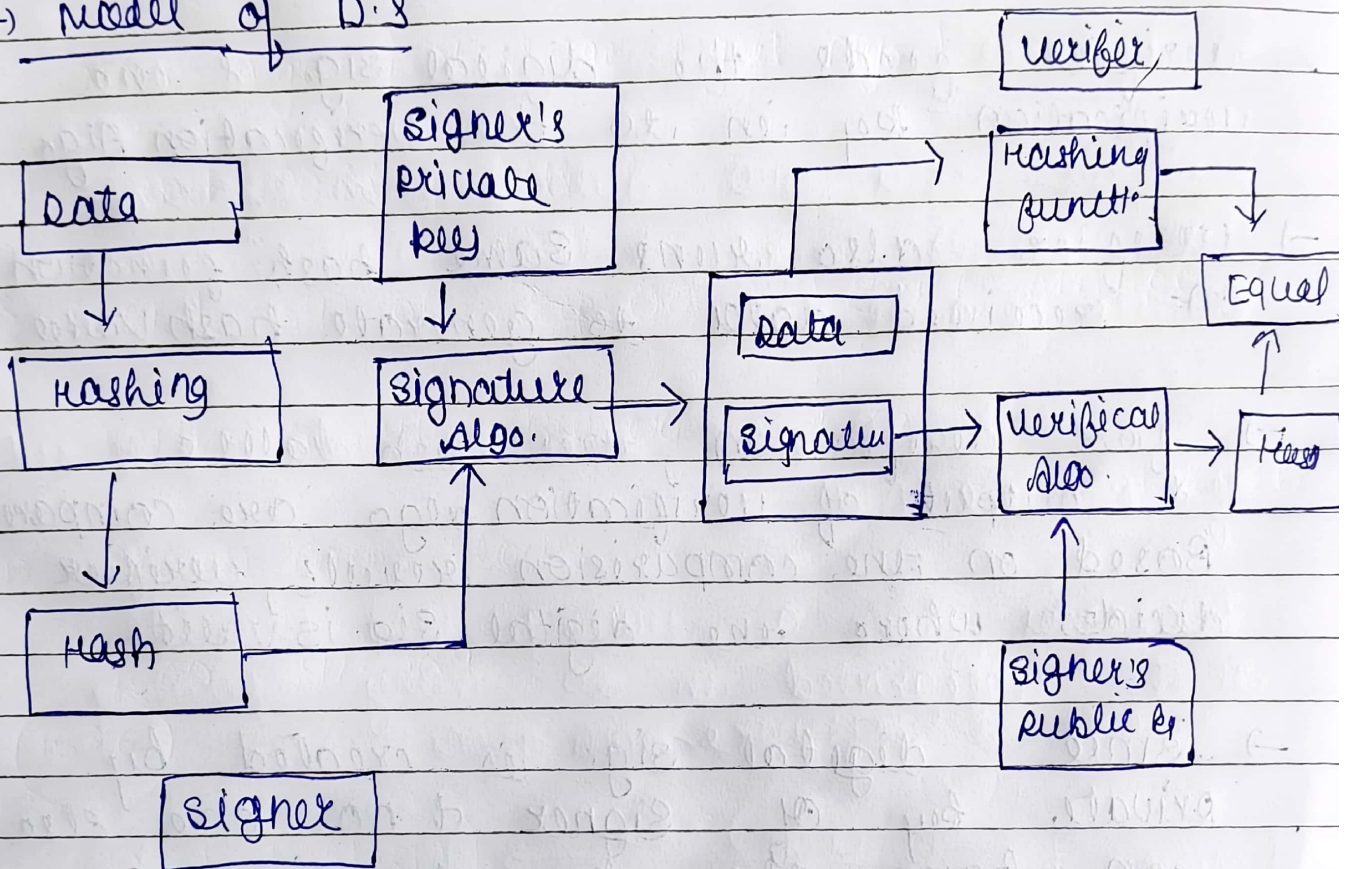
$$H(B, C, D) \rightarrow B \oplus C \oplus D$$

$$I(B, C, D) \rightarrow C \oplus (B \vee \neg D)$$

⇒ Digital Signature

→ D.S is a cryptographic value that is calculated from the data & a ~~secret~~ secret key only known by the signer.

→ Model of D.S



→ The following points explain the entire process in detail.

→ Each person adopting this scheme has public-private key pair.

→ The private key used for signing is referred to as the signature key & public key as the verification key.

- signer feeds data to the hash function & generates hash of data.
- Hash value & signature key are then fed to the signature Algo. which produces the d.s of the on given hash.
- verifier feeds the digital sig. & the verification key in to the Verification Algo.
- verifier also runs same hash function on received data to generate hash value.
- for verification, this hash value & output of verification algo. are compared. Based on the comparison result, verifier decides where the digital sig. is valid.
- since digital sig. is created by private key of signer & no. one else can have this key, the signer cannot repudiate signing the data.

unit 14

SSL [~~ssl or secure s~~]

X.509 →

X.509 is a standard that defines the format of public key certificate. These certificates are used in various internet protocols, including TLS/SSL (The foundation of HTTPS).

X.509

- version → usually V3.
- serial no. → unique identifier of the certificate.
- Signature Algo. → Algo. used to sign the certificate [SHA 256 and RSA].
- Issuer → The entity that issued the certificate.
- validity period.
- subject public key info.
- extensions.
- signatures.

→ use cases →

→ HTTPS / SSL (Secure Socket Layer).

→ Email encryption.

→ Code signing

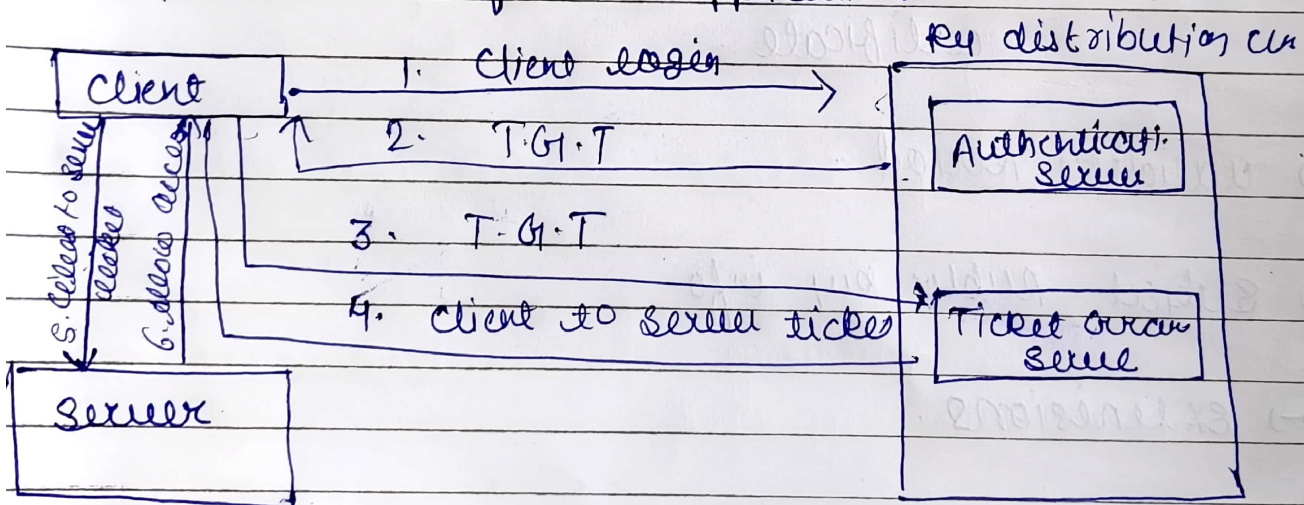
→ VPNs & Secure N/w Access.

⇒ Kerberos

Kerberos is a n/w authentication protocol that verifies user identities & authentication service request.

→ It's used to keep unauthorized users out of private N/w.

→ Kerberos is a widely used protocol that's integrated in to many operating systems & software applications.



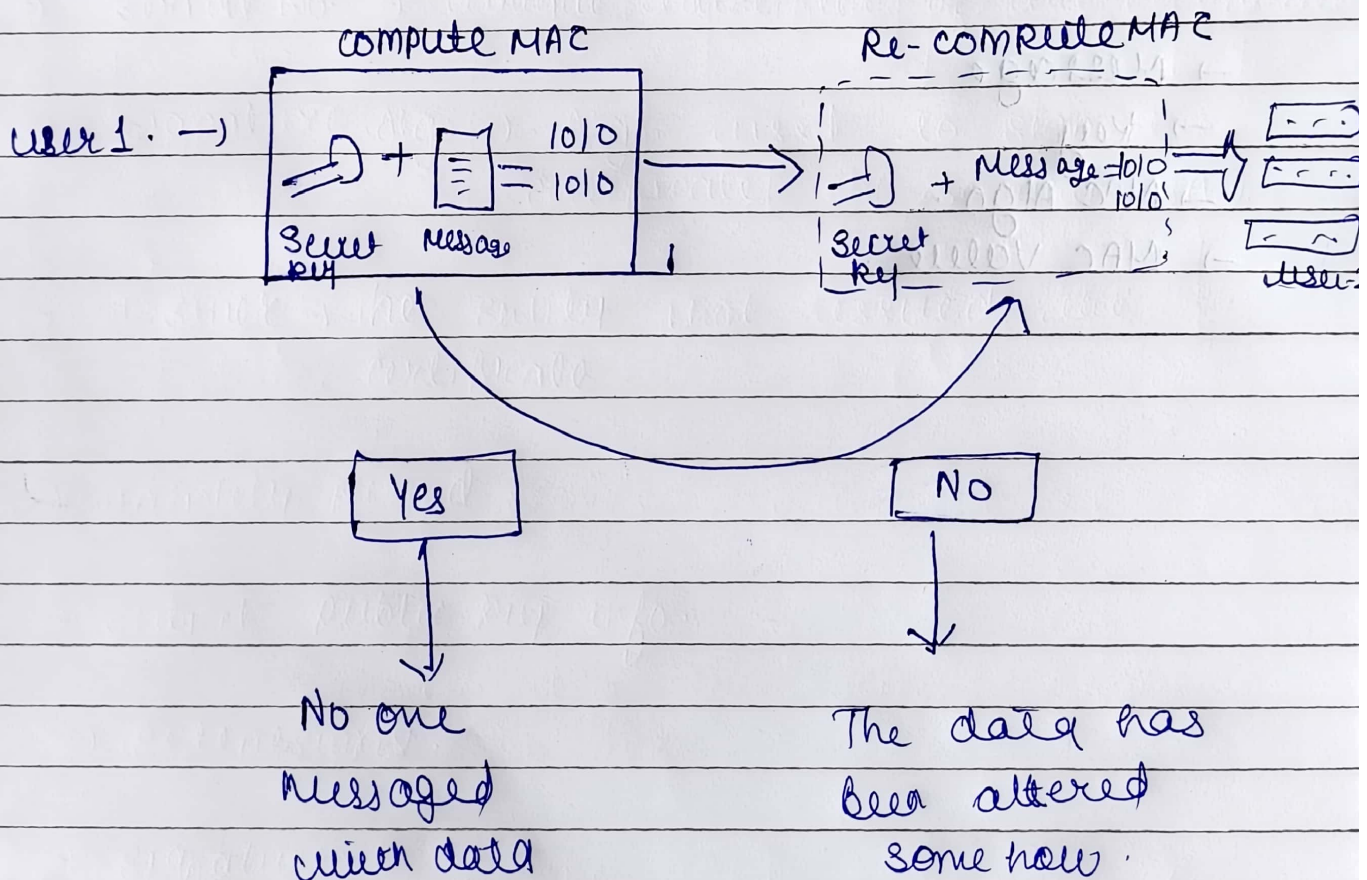
→ key components

→ KDC [key distribution center]. (KDC) →
A trusted third party that authenticates
users & issues tickets.

→ principal → A unique identity in a Kerberos
system

→ Ticket → A secure way to verify a
user's identity.

→ Message Authentication code.



→ MAC stands for Message Authentication Code. Here in MAC sender & receiver share same key, where sender generates a fixed size output called cryptographic checksum or Message Authentication code & appends it to the original message.

→ On receiver's side, receiver also generates the code & compares it with what he/she received thus ensuring the originality of message.

→ Components

→ Message

→ Key

→ MAC Algo.

→ MAC Value